BHARAT INSTITUTE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institution - UGC, GOVT. of INDIA) NAAC
Accredited with 'A' Grade, NBA Accredited for UG
Programs: CSE, ECE
Approved by AICTE, New Delhi, Affiliated to JNTUH

LITERATURE SURVEY AND RELATED WORK ANALYSIS

PROBLEM STATEMENT:

- Farmers face many challenges such as crop diseases, unpredictable weather, poor soil quality, incorrect fertilizer usage, and fluctuating market prices.
- Most existing agricultural systems provide only one service, such as disease detection, weather forecasting, or crop recommendation.
- Farmers need to use multiple applications to get complete farming information, which is time-consuming and inconvenient.
- Many existing systems provide information only in English, making them difficult for local farmers to understand.
- Farmers often do not receive personalized recommendations based on their crop, soil, weather, and location.
- Existing systems do not integrate Generative AI and Multi-Agent AI to provide complete farming guidance.
- There is no single web-based platform that combines crop disease detection, weather updates, soil and fertilizer recommendations, market price information, and an AI chatbot.
- Farmers need a simple, intelligent, and multilingual farming assistant that provides accurate and easy-to-understand recommendations

.**MOTIVATION:**

- Agriculture is the backbone of the economy, and farmers need timely and accurate guidance to improve crop productivity.
- Farmers often face problems such as crop diseases, unpredictable weather, poor soil conditions, incorrect fertilizer usage, and changing market prices.
- Most existing agricultural applications solve only one problem, forcing farmers to use multiple platforms for different services.
- Many farmers, especially in rural areas, have limited access to agricultural experts and do not receive timely advice.
- Existing systems often provide information only in English and lack support for local languages, making them difficult for many farmers to use.
- Recent advancements in Artificial Intelligence (AI), Multi-Agent Systems, and Generative AI make it possible to provide intelligent, personalized, and easy-to-understand farming recommendations.
- Developing a single web-based platform that integrates disease detection, weather forecasting, soil and fertilizer recommendations, market prices, and an AI chatbot can make farming decisions easier and more effective.
- This project aims to improve farmers' productivity, reduce crop losses, increase income, and promote smart and sustainable agriculture through AI technology**.**

OBJECTIVES:

- Objective 1:
  Develop a web-based AI Multi-Agent Farming Assistant for farmers. Detect crop diseases using Deep Learning and image analysis.
- Objective 2:
  Provide personalized farming advice using Generative AI.
  Integrate weather forecasting, soil recommendations, and market price information.
- Objective 3
  Support multilingual communication and voice interaction.
  Improve agricultural productivity through intelligent decision support.

LITERATURE SURVEY:

S. No	Paper Title	Authors	Year	Journal / Conference	Methodology	Advantages	Limitations
1	AgroAskAI: A Multi-Agentic AI Framework for Supporting Smallholder Farmers' Enquiries Globally	W. Zhang et al.	2026	AAAI Conference (Scopus)	Multi-Agent AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)	Personalized recommendations, multilingual support, farmer decision support	No disease detection, no soil analysis, no market price integration
2	Generative AI as an Interface Between Farmers and AI Algorithms: Opportunities and Challenges	W. Zhang and A. Yadav	2025	Springer – Journal of the Indian Institute of Science	Generative AI, LLMs, Conversational AI	Easy interaction, supports low-literacy farmers, AI-based decision support	No complete farming platform, no weather or disease modules
3	Deep Learning in Agriculture: A Survey	A. Kamilaris and F. X. Prenafeta-Boldú	2018	Elsevier – Computers and Electronics in Agriculture	Deep Learning, CNN	Comprehensive review of AI in agriculture, high prediction accuracy	Does not provide an integrated farming solution
4	Machine Learning in Agriculture: A Review	K. G. Liakos et al.	2018	Sensors (Scopus Indexed)	Machine Learning, Data Analytics	Reviews ML techniques for precision farming	Mostly theoretical, lacks Generative AI integration
5	Using Deep Learning for Image-Based Plant Disease Detection	S. P. Mohanty, D. P. Hughes, M. Salathé	2016	Frontiers in Plant Science	CNN, Image Classification	Accurate plant disease detection	Focuses only on disease detection

6	Big Data in Smart Farming – A Review	S. Wolfert et al.	2017	Elsevier – Agricultural Systems	Big Data Analytics, Internet of Things (IoT)	Explains smart farming technologies	No conversational AI or decision-support system
7	The Role of Large Language Models in Agriculture: Harvesting the Future with LLM Intelligence	T. A. Shaikh et al.	2024	Progress in Artificial Intelligence (Springer)	Large Language Models (LLMs)	Explains agricultural applications of LLMs	No implementation of a complete multi-agent farming system

RELATED WORK ANALYSIS:

Existing Method	Technology Used	Drawbacks	Proposed Improvement
AgroAskAI – Multi-Agent Farming Assistant	Multi-Agent AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Weather APIs	No crop disease detection, no soil analysis, no fertilizer recommendation, no market price information, no farmer dashboard	Integrate crop disease detection, soil analysis, fertilizer recommendation, live market prices, weather forecasting, multilingual support, voice assistance, and a farmer dashboard in one website.
Generative AI as an Interface Between Farmers and AI Algorithms	Generative AI, Large Language Models (LLMs), Natural Language Processing (NLP)	Provides only conversational support, no disease detection, no crop recommendation, no weather forecasting, no market price module	Develop a complete AI-powered farming website with Generative AI chatbot, disease detection, crop recommendation, weather updates, fertilizer suggestions, and market prices.
Plant Disease Detection System	Deep Learning, CNN, Image Processing	Detects only crop diseases and does not provide complete farming guidance	Add treatment suggestions, weather-based recommendations, soil analysis, and Generative AI explanations.
Crop Recommendation System	Machine Learning (Random Forest, Decision Tree, SVM)	Recommends crops only; does not consider diseases, market prices, or weather updates	Combine crop recommendation with disease detection, weather forecasting, soil analysis, and market price information.

GAP ANALYSIS:

- Existing systems focus on only one farming service, such as crop disease detection, weather forecasting, or crop recommendation.
- There is no single platform that integrates all farming services in one website.
- Most existing systems do not use Multi-Agent AI to handle multiple farming tasks simultaneously.
- Existing applications provide limited personalized recommendations and do not combine crop, soil, weather, and market information.
- Many systems do not support multilingual communication, making them difficult for local farmers to use.
- Existing chatbots provide general or predefined answers instead of intelligent, context-aware recommendations.
- Most systems do not support voice interaction, which is useful for farmers who are not comfortable typing.
- Existing solutions often lack real-time market price integration, making it difficult for farmers to decide when and where to sell their crops.
- Very few systems combine Generative AI with crop disease detection, weather forecasting, soil analysis, fertilizer recommendation, and market price information in a single platform.
- There is a need for a complete AI-powered web-based farming assistant that provides personalized, accurate, multilingual, and easy-to-understand recommendations.

PROPOSED SOLUTION:

- Develop a web-based AI Multi-Agent Farming Assistant to provide all farming services on a single platform.
- Use Multiple AI Agents to perform different tasks such as crop disease detection, weather forecasting, soil analysis, fertilizer recommendation, and market price updates.
- Integrate Generative AI to provide personalized and easy-to-understand farming advice.
- Allow farmers to upload crop images for automatic disease detection and treatment suggestions.
- Provide real-time weather information and live market prices to support better farming decisions.
- Support multilingual and voice-based interaction for easy communication.
- Help farmers improve crop productivity, reduce crop losses, and increase income through intelligent recommendations.

NOVELTY OF THE PROJECT:

- Integrates Multiple AI Agents into a single web-based farming platform.
- Combines Generative AI with crop disease detection, weather forecasting, soil analysis, fertilizer recommendation, and market price updates.
- Provides personalized farming recommendations based on real-time data and farmer queries.
- Supports multilingual and voice-based interaction for easy communication.
- Allows crop image upload for AI-based disease detection and treatment suggestions.
- Offers a complete farming solution instead of separate applications for different services.
- Helps farmers make faster, smarter, and more accurate decisions to improve crop productivity and reduce losses.

SOFTWARE AND HARDWARE REQUIREMENT:

SOFTWARE:

- Windows 10/11
- Python
- Flask
- React.js
- MongoDB
- TensorFlow
- OpenCV
- Google Gemini API
- Visual Studio Code
- Git
- Postman

HARDWARE:

- Processor: Intel Core i5 or above
- RAM: 8 GB (Minimum), 16 GB (Recommended)
- Storage: 512 GB SSD
- Internet Connection
- Webcam / Mobile Camera (Optional)
- Microphone (Optional
- Monitor/Display

DATASET DETAILS:

Dataset Name	Source	Size
PlantVillage data set	kaggle	~1.3 GB (54,000+ images)
Crop Recommendation Dataset	kaggle	~220 KB (2,200 records)
Fertilizer Prediction Dataset	kaggle	~25KB
Crop yield prediction Dataset	kaggle	5-10MB
Agmarknet market price dataset	Data.Gov.in	Varies

EXPECTED OUTCOMES:

- Accurate crop disease detection using AI.
- Personalized farming recommendations using Generative AI.
- Real-time weather forecasting and alerts.
- Suitable crop and fertilizer recommendations.
- Live market price information for better selling decisions.
- Easy-to-use multilingual and voice-enabled web platform.
- Improved farmer decision-making and reduced crop losses.
- Increased crop productivity and farmers' income.

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REVIEW COMMITTEE REMARKS:

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Signature of the Student: _____